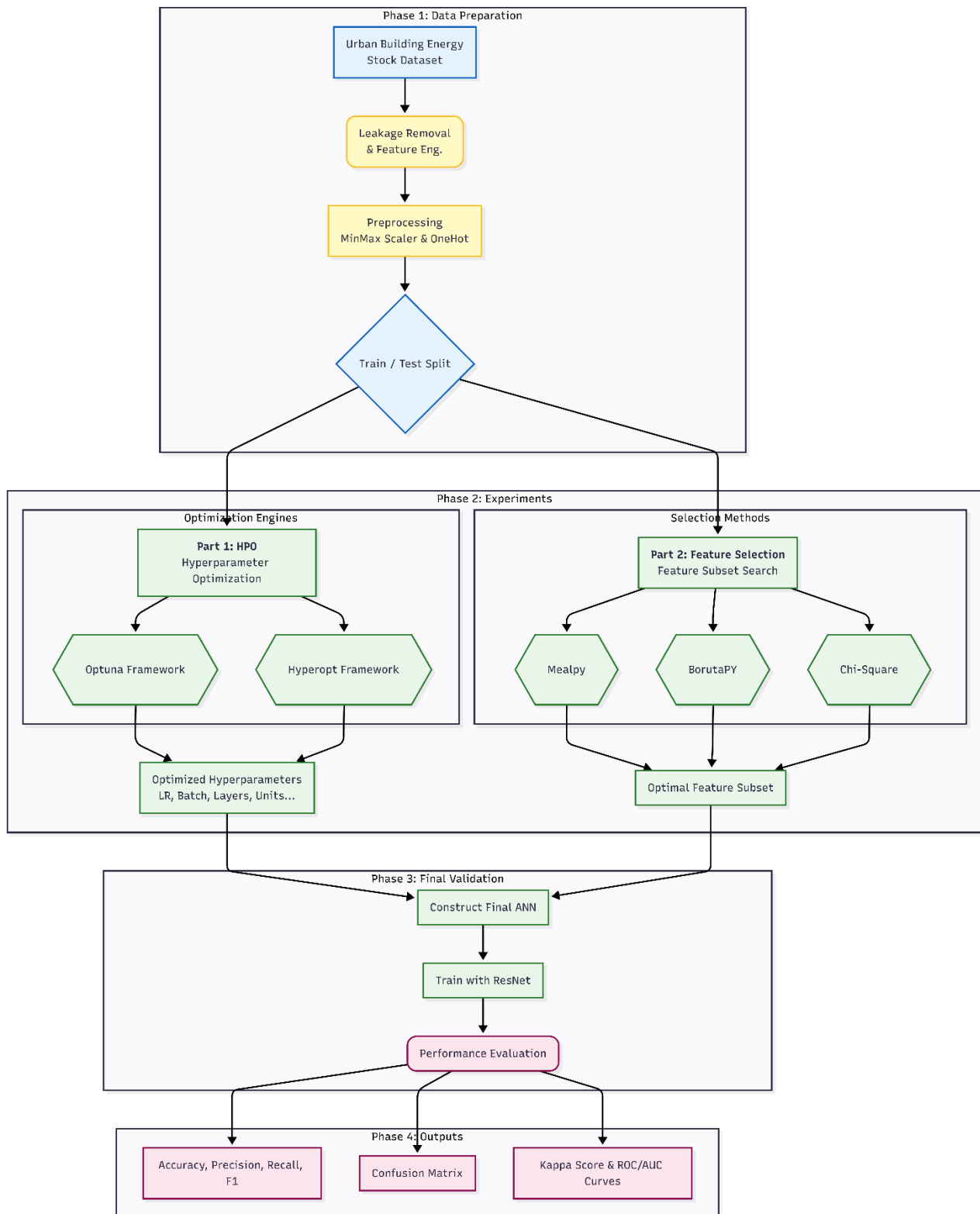


CSE443 – CE471 AI For Intelligent Built Environment Systems
Fall 2025 - Assignment#2

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A graphical abstract for methodology.



Part 1 - Apply Hyperparameter optimization on ANN

Selected hyperparameter optimization technique 1: **Optuna**

Selected hyperparameter optimization technique 2: **Hyperopt**

Hyperparameter Space used:

Hyperparameter	Search Space
Learning Rate	Continuous (Log): [0.0001, 0.01]
Batch Size	256, 512, 1024
Number of Layers	2,3,4
Number of Units/Neurons	Integer: [64, 1024] (step=64)
Dropout Rate	Float: [0.0, 0.5] (step=0.1)
Activation Function	ReLU, Swish
Optimizer	Adam, RMSProp

Comparison:

Technique	Accuracy	Precision	Recall	F1-Score	Training Runtime	Optimization Runtime	Cohen's Kappa
Optuna	0.9030	0.9032	0.9030	0.9030	28.23 min	22.98 min	0.8786
Hyperopt	0.9169	0.9171	0.9169	0.9170	13.83 min	26.04 min	0.8958
Algorithm performance without HPO	0.9002	0.9003	0.9002	0.9001	N/A	N/A	0.8749

For the best case

```
Hyperopt (TPE) Optimizasyonu Başlıyor...
100%|██████████| 30/30 [26:02<00:00, 52.08s/trial, best loss: -0.8390713334083557]

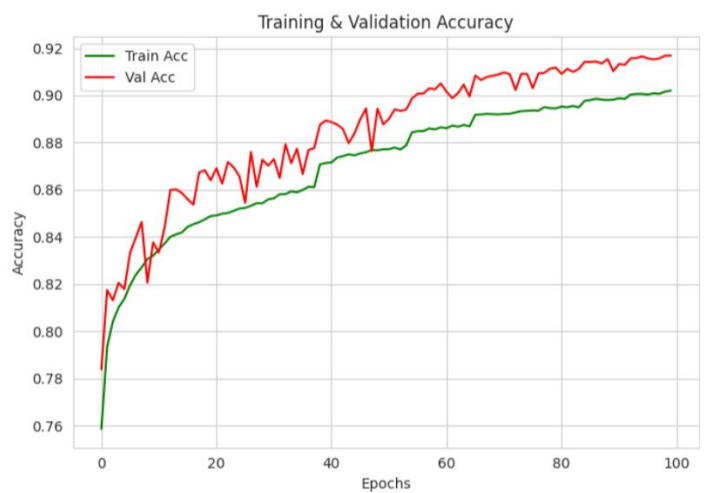
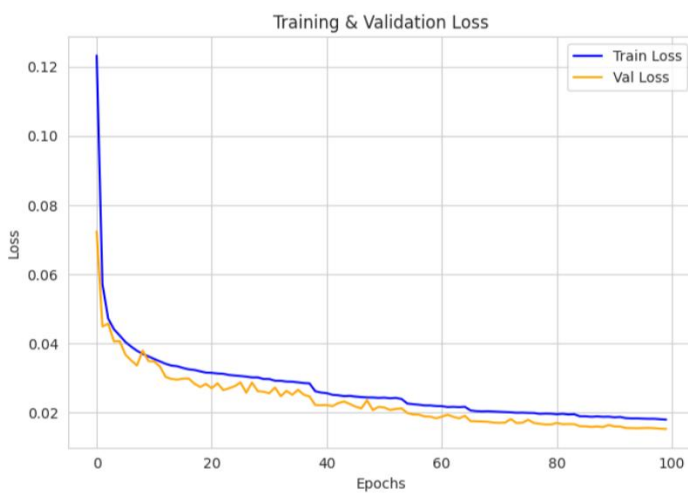
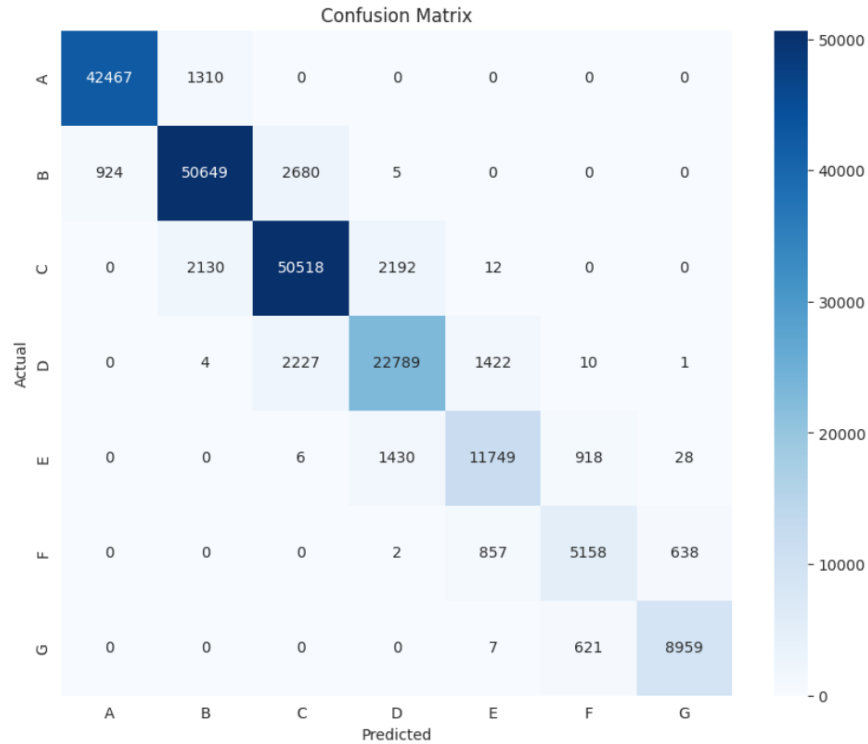
Hyperopt Süresi: 26.04 dakika

KAZANAN PARAMETRELER:
{'learning_rate': np.float64(0.000674289500257529), 'batch_size': 512, 'n_layers': 3, 'units': 384, 'activation': 'relu', 'optimizer': 'adam', 'dropout': np.float64(0.005684569598789768)}
```

- Draw a table of accuracy, precision, recall, F1 results of each class.

--- Classification Report ---				
	precision	recall	f1-score	support
A	0.9787	0.9701	0.9744	43777
B	0.9363	0.9335	0.9349	54258
C	0.9114	0.9210	0.9162	54852
D	0.8626	0.8615	0.8621	26453
E	0.8364	0.8314	0.8339	14131
F	0.7690	0.7751	0.7720	6655
G	0.9307	0.9345	0.9326	9587
accuracy			0.9169	209713
macro avg	0.8893	0.8896	0.8894	209713
weighted avg	0.9171	0.9169	0.9170	209713

- Represent the confusion matrix, training/validation loss and training/validation accuracy graphs



- Show the validation and test results by commenting on them.

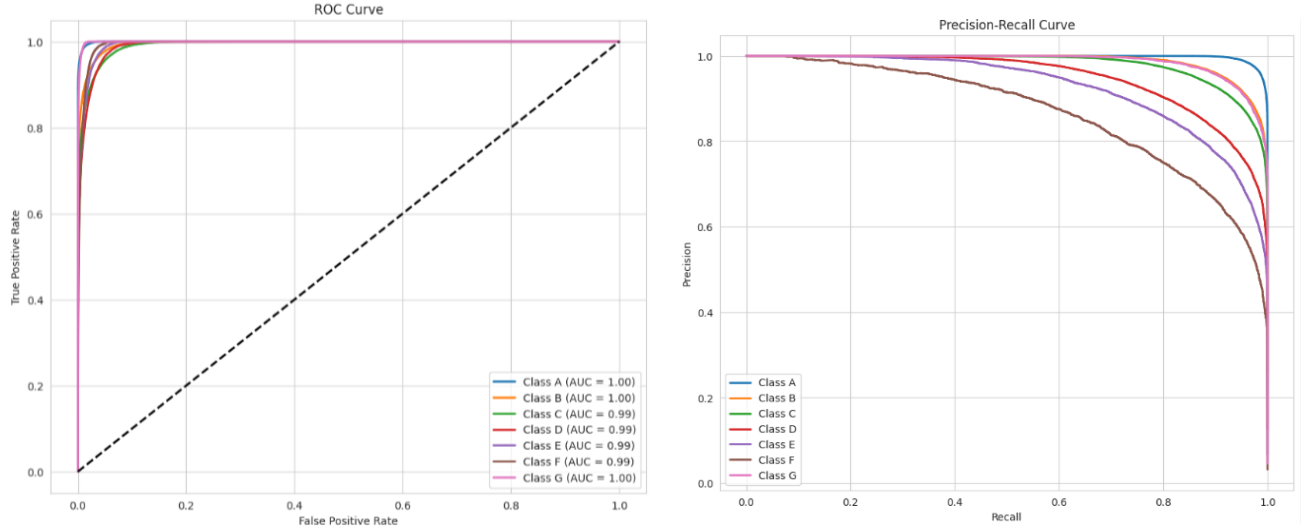
```
Epoch 100/100
1639/1639 — 8s 5ms/step - accuracy: 0.9021 - loss: 0.0180 - val_accuracy: 0.9169 - val_loss: 0.0153 - learning_rate: 1.0536e-05
```

```
accuracy 0.9169 209713
```

- Calculate Cohen's Kappa score and print as output

```
Cohen's Kappa Score: 0.8958
HPO Runtime: 26.04 min
Training Runtime: 13.83 min
```

- Draw precision recall and AUC-ROC curve



Part 2

1)Meta-heuristic algorithm: Genetic Algorithm

2)BorutaPY

3)Filter based feature selection technique: Chi-square

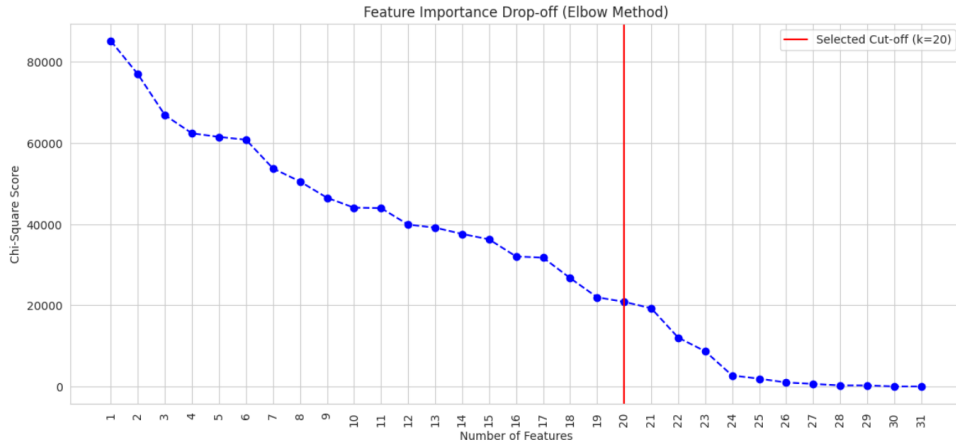
Feature Importance Comparison Table (Top 15):

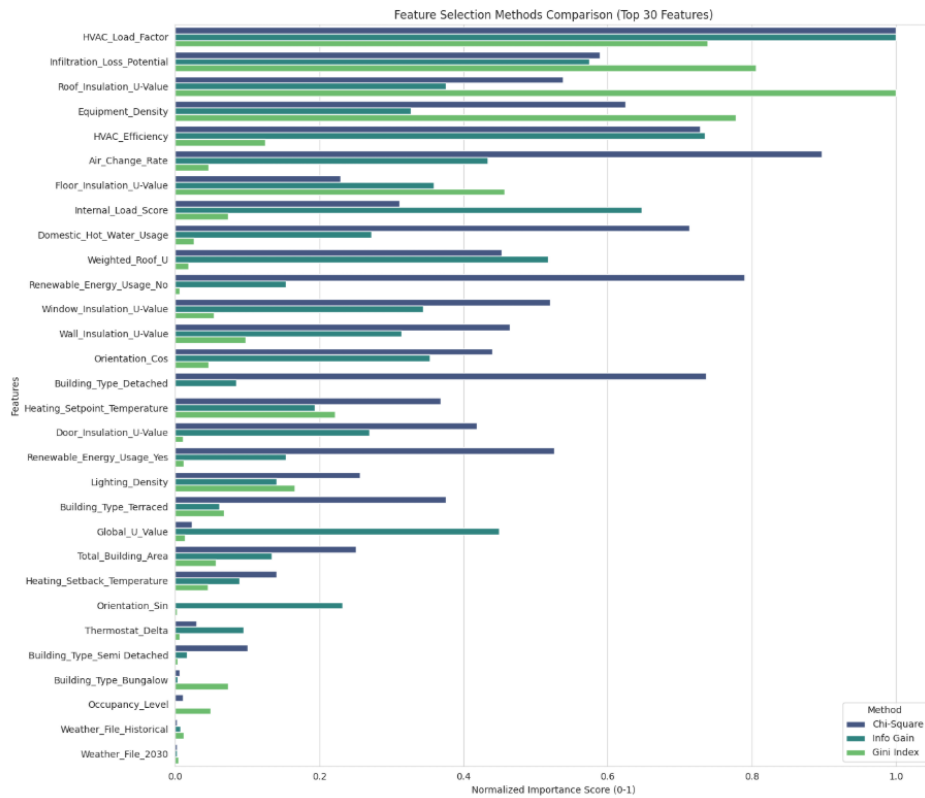
Feature	Chi-Square	Info Gain	Gini Index	Mean Score
HVAC_Load_Factor	1.000000	1.000000	0.738645	0.912882
Infiltration_Loss_Potential	0.588817	0.574048	0.805987	0.656284
Roof_Insulation_U-Value	0.538345	0.375739	1.000000	0.638028
Equipment_Density	0.624266	0.327518	0.777226	0.576337
HVAC_Efficiency	0.728193	0.735170	0.124808	0.529391
Air_Change_Rate	0.896884	0.433415	0.046447	0.458915
Floor_Insulation_U-Value	0.229723	0.358606	0.457150	0.348493
Internal_Load_Score	0.311454	0.647308	0.073907	0.344223
Domestic_Hot_Water_Usage	0.712996	0.271775	0.026327	0.337032
Weighted_Roof_U	0.452923	0.517235	0.018373	0.329510
Renewable_Energy_Usage_No	0.790203	0.154092	0.006846	0.317047
Window_Insulation_U-Value	0.520365	0.344085	0.053609	0.306020
Wall_Insulation_U-Value	0.464365	0.314171	0.097698	0.292078
Orientation_Cos	0.439860	0.353134	0.046305	0.279766
Building_Type_Detached	0.736860	0.085021	0.000000	0.273960

ANALİZ SONUCU:

Üç yöntemin ortak kararına göre en önemli özellik: 'HVAC_Load_Factor'

Özellik Önem Dağılımı Analiz Ediliyor...





In this study, three filter-based feature selection methods—Chi-Square, Information Gain (Mutual Information), and Gini Index—were employed to evaluate the importance of input features. Since these methods generate importance scores on different numerical scales, all scores were normalized to the [0,1] range using Min–Max normalization to ensure fair comparison.

The results showed that **HVAC_Load_Factor** consistently ranked as the most important feature across all three methods. Among the evaluated techniques, **Chi-Square** was selected as the primary filter-based feature selection method, as it effectively captured the statistical dependency between the input features and the categorical target variable (*Simple_Building_Energy_Rating*), while also producing physically interpretable and stable results aligned with building energy principles.

Furthermore, the Chi-Square–based feature importance drop-off graph (Elbow Method), presented in Figure X, reveals a clear elbow point after approximately the first 20 features. Beyond this point, the contribution of additional features becomes marginal. Based on this observation, the top 20 features were selected to balance model complexity and information preservation. Overall, supported by both cross-method consistency and elbow analysis, Chi-Square was adopted as the final filter-based feature selection approach in this study.

Comparison:

Technique	Accuracy	Precision	Recall	F1-Score	Training Runtime	Selection Runtime	Cohen's Kappa
Genetic Algorithm	0.8257	0.8251	0.8257	0.8252	29.44 min	45 min	0.7813
BorutaPy	0.8796	0.8800	0.8796	0.8797	27.40 min	1.14 min	0.8491
Chi-Square	0.8782	0.8784	0.8782	0.8782	27.21 min	45 sec	0.8473
Result without feature selection	0.9030	0.9032	0.9030	0.9030	N/A	N/A	0.8786

List the features selected in best case.

- Draw a table of accuracy, precision, recall, F1 results of each class.

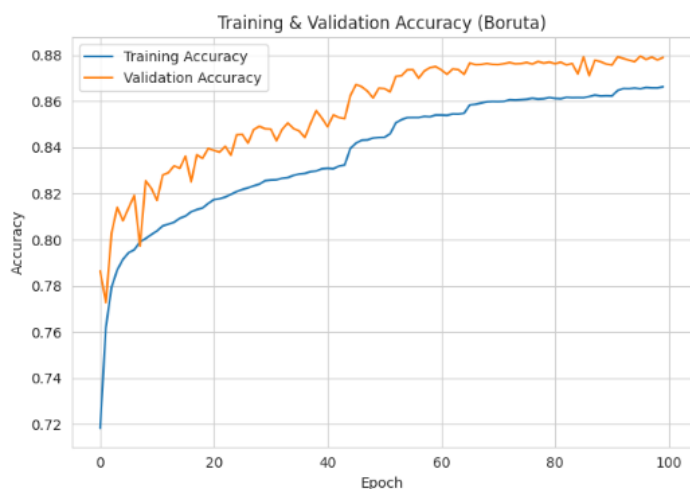
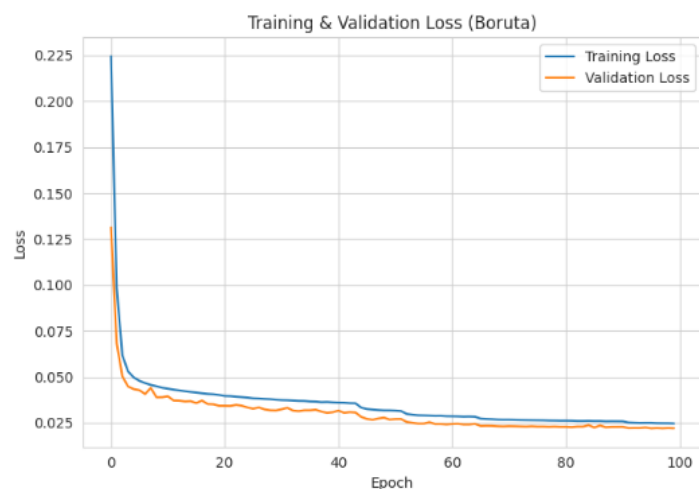
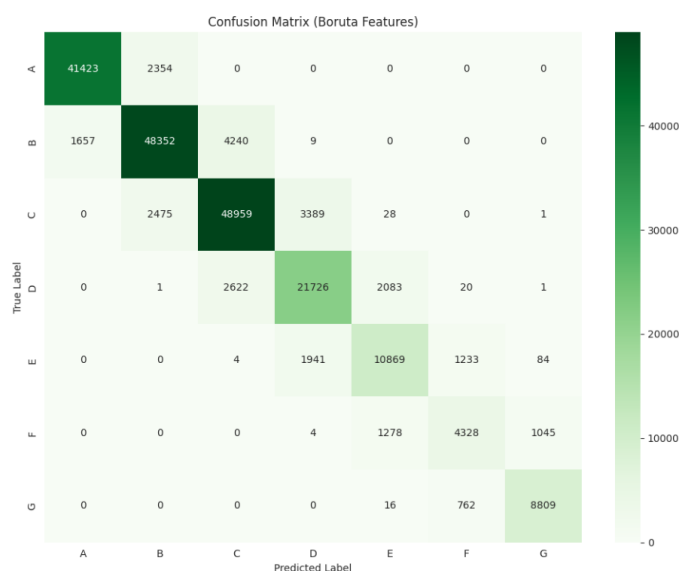
```

--- Classification Report (Boruta Features) ---

```

	precision	recall	f1-score	support
A	0.9615	0.9462	0.9538	43777
B	0.9092	0.8911	0.9001	54258
C	0.8770	0.8926	0.8847	54852
D	0.8026	0.8213	0.8119	26453
E	0.7615	0.7692	0.7653	14131
F	0.6823	0.6503	0.6659	6655
G	0.8862	0.9188	0.9022	9587
accuracy			0.8796	209713
macro avg	0.8400	0.8414	0.8406	209713
weighted avg	0.8800	0.8796	0.8797	209713

- Represent the confusion matrix, training/validation loss and training/validation accuracy graphs.



- Show the validation and test results by commenting on them.

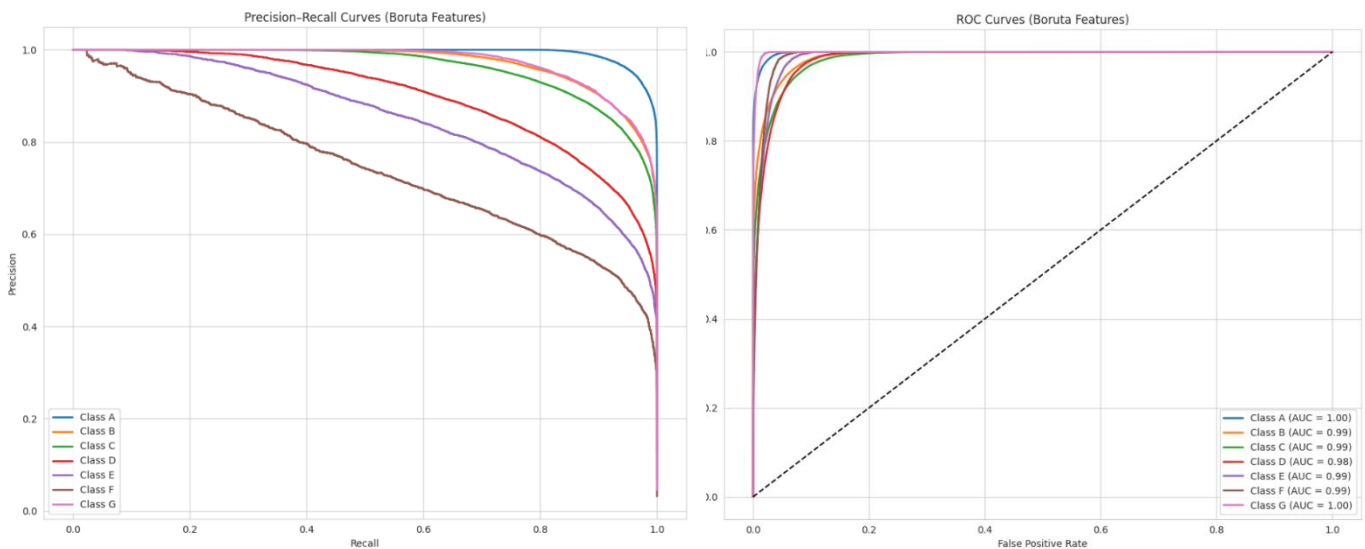
```
Epoch 96/100
3277/3277 ————— 16s 5ms/step - accuracy: 0.8660 - loss: 0.0249 - val_accuracy: 0.8796 - val_loss: 0.0220 - learning_rate: 1.2832e-05
Epoch 97/100
3277/3277 ————— 17s 5ms/step - accuracy: 0.8665 - loss: 0.0247 - val_accuracy: 0.8782 - val_loss: 0.0222 - learning_rate: 1.2832e-05
Epoch 98/100
3277/3277 ————— 16s 5ms/step - accuracy: 0.8656 - loss: 0.0248 - val_accuracy: 0.8792 - val_loss: 0.0220 - learning_rate: 1.2832e-05
Epoch 99/100
3277/3277 ————— 16s 5ms/step - accuracy: 0.8660 - loss: 0.0247 - val_accuracy: 0.8780 - val_loss: 0.0222 - learning_rate: 1.2832e-05
Epoch 100/100
3277/3277 ————— 17s 5ms/step - accuracy: 0.8655 - loss: 0.0247 - val_accuracy: 0.8790 - val_loss: 0.0221 - learning_rate: 1.2832e-05
Restoring model weights from the end of the best epoch: 96.
Eğitim Süresi: 27.40 dk
```

accuracy 0.8796 209713

- Calculate Cohen's Kappa score and print as output

Test Set Accuracy Rate: %87.96
Cohen's Kappa Score: 0.8491

- Draw precision recall and AUC-ROC curve



Discussion

This study demonstrates that artificial intelligence models, when combined with building-physics-based features, can accurately predict residential building energy performance without requiring time-consuming physical inspections or simulation-based evaluations. Traditional Building Energy Rating (BER) assessments rely on on-site audits, material testing (e.g., insulation thickness, window types), and simulation tools such as EnergyPlus. These approaches are costly, slow, and impractical for large urban building stocks. In contrast, our ANN-based framework predicts the energy class of a building (A–G) using only a limited number of fundamental physical descriptors such as U-values, window-to-wall ratio, HVAC efficiency, and air change rate, achieving up to 91.69% classification accuracy with Hyperopt optimization.

The results of hyperparameter optimization clearly show that automated tuning techniques significantly improve model performance. While the baseline ANN achieved an accuracy of approximately 90.02%, the use of Hyperopt increased accuracy to 91.69%, with corresponding improvements in precision, recall, and F1-score. This indicates that the learning capacity of deep neural networks in building energy modeling is highly sensitive to architectural and training parameters such as learning rate, number of neurons, and dropout rate. Therefore, data-driven optimization strategies are essential for extracting maximum predictive power from large-scale building datasets.

From a civil engineering perspective, the most important finding of this study is the dominant role of building envelope-related features. Variables such as Global U-Value, weighted roof U-value, and infiltration loss potential were consistently selected or ranked highly during feature selection experiments. This aligns strongly with building physics theory, which states that heat loss through walls, roofs, windows, and air leakage is the primary driver of space heating demand. The success of the model therefore provides quantitative evidence that thermal resistance and envelope design are the key determinants of building energy performance, validating long-established principles in building engineering.

The feature selection experiments further highlight the engineering interpretation of the results. Although BorutaPy and Chi-square methods were able to eliminate less informative variables, the highest performance was achieved when the full physics-informed feature set was retained. This suggests that building energy behavior is governed by complex nonlinear interactions between insulation, geometry, HVAC efficiency, and occupancy-related loads. Removing these interacting parameters, even if they appear weak individually, reduces the model's ability to represent the true thermodynamic behavior of buildings.

Another critical implication of this work is its relevance to retrofitting and urban energy planning. The model reliably identifies poorly performing buildings (Class F and G), which are the primary targets for energy-efficient renovation strategies such as façade insulation, roof retrofitting, and window replacement. In a real-world urban context, this enables engineers and municipalities to prioritize investments by automatically scanning large building stocks and selecting those with the highest potential for energy savings. Instead of performing expensive audits on every building, AI-based screening can drastically reduce both time and cost.

Finally, the high accuracy and robustness of the proposed system make it suitable for Smart City and Digital Twin applications. By integrating this ANN into urban-scale digital models, it becomes possible to estimate the energy demand and retrofit potential of thousands of buildings simultaneously. This transforms building energy analysis from a building-by-building manual process into a scalable, city-wide, data-driven decision support system.

In conclusion, this study shows that artificial neural networks enriched with physics-based features can serve as a powerful alternative to traditional building energy assessment methods. The proposed approach enables fast, accurate, and scalable evaluation of building energy performance, supporting sustainable retrofit planning and the development of intelligent built environments.